

Today will cover:

1. Local variable
2. Instance variable
3. Static variable
4. Static block
5. Access modifiers
6. Non-access modifiers
7. Final
8. Abstract class
9. Interface
10. String
11. StringBuffer
12. StringBuilder

If we create 100 student objects, should every student have sperate name ?

Yes

If all students belong to the same institute should every object store a separate institute name ?

Institute will be same for all student : NO

Type of variable(area) in java

```
public class Student { no usages
    void cal() no usages
    {
        int marks=85;
        String result="pass";

        System.out.println(marks);
        System.out.println(result);
    }
}
```

Marks and result are local variable
They exist only inside cal()

a

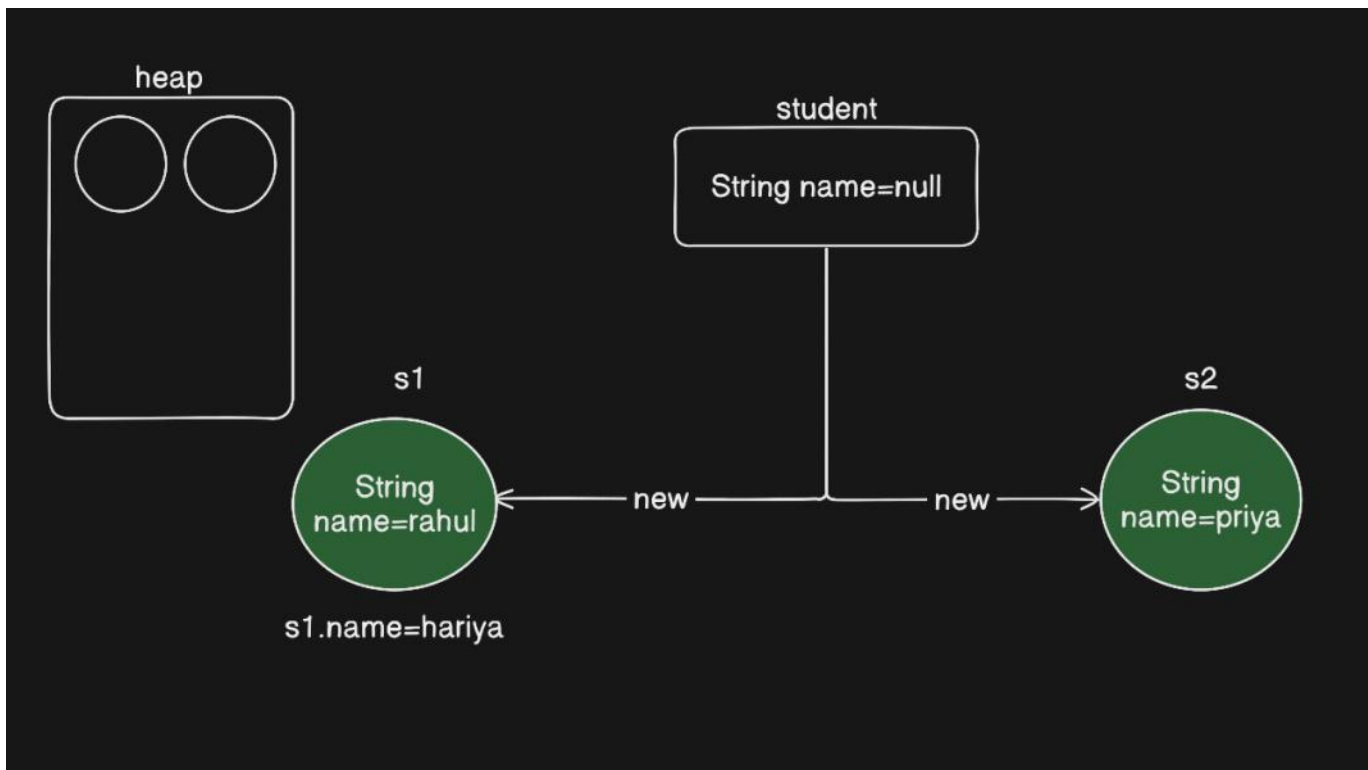
Important points:

1. Local variable are created when method/block start
2. Local variable are destroyed when method/block ends
3. Local variable cannot be used outside their method/block
4. Local variable do not get default values
5. We must initialize Local variable before using them

Instance variable:

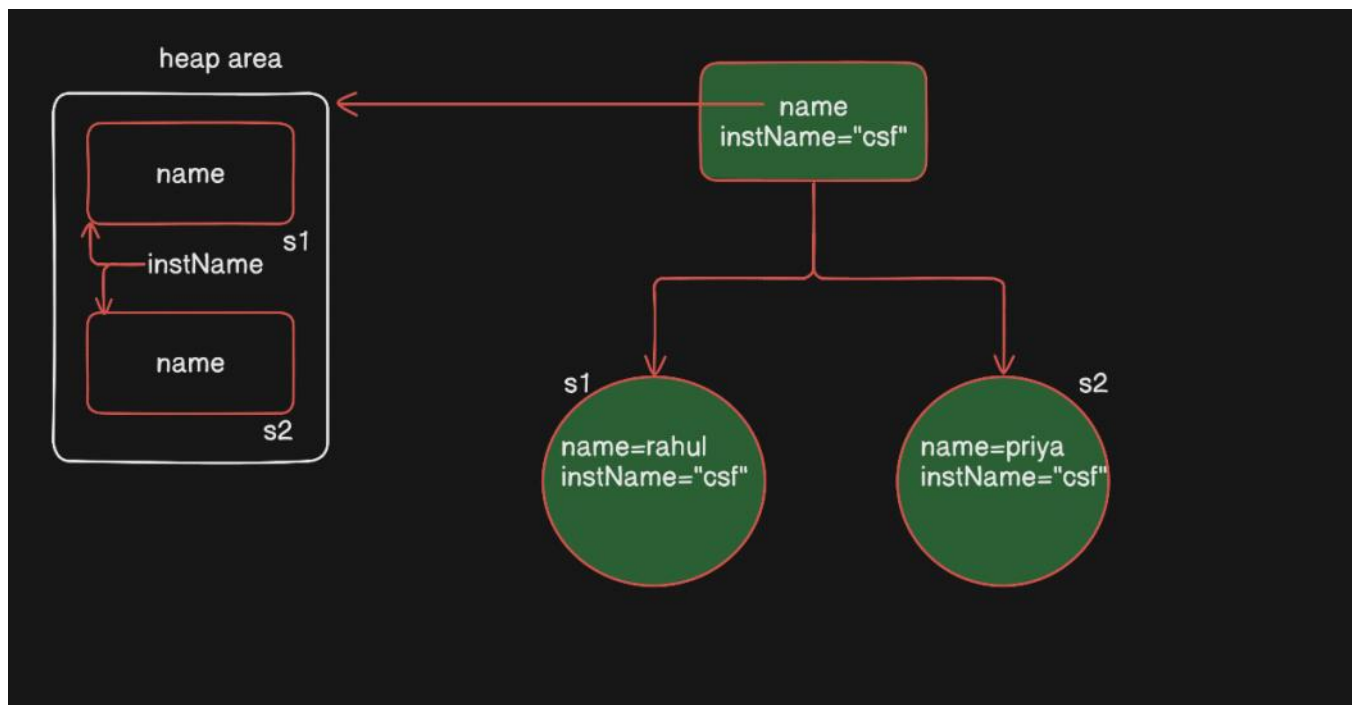
An instance variable is declared inside a class but outside any method, constructor or block:
Instance variable = object-level variable

```
public class Student { no usages
    String name; no usages
    int age; no usages
    int marks; no usages
}
```



Static variable:

Static variable = class-level variable shared by all objects



Instance variable Vs Static variable

Point	Instance variable	Static variable
Belongs to	Object	Class
Memory copy	Separate copy for each object	Single copy
Access	Reference variable	Class name

```

1  public class Course {
2
3      String name="Spark 6.0"; no usages
4      static String instName="CFS"; no usages
5
6  public static void main(String[] args) {
7
8
9
10
11
12
13
14
15
16

```

Static area

Static area : in static are we can only use static variables
Non static area: we can use both instance and static variable

Q1: why main method is static ?

JVM can call main method without creating object of the class.

Static block

A static block is used to initialize static data.

Access Modifiers

Access modifiers control visibility.

Java has four access levels:

-
1. private
 2. default
 3. protected
 4. public

Important:

If we do not write any access modifier, it is default access.

Modifier	Same Class	Same Package	Child class Different package	Anywhere
Private	Yes	No	No	No
Default	Yes	Yes (child class/other class)	no	No
Protected	Yes	Yes	Yes	No
Public	Yes	Yes	Yes	yes

Private:

Private variable, method will not be available for inheritance and only accessible inside the class.

Default:

Default member can be accessed inside the same package

```

3
4  class Student { no usages
5
6      String name; 1 usage
7
8      void show() no usages
9      {
10         System.out.println(name);
11     }
12 }
13

```

Student class, name variable and show method have default access.

Java does not allow protected and private on top - level classes.

Non-Access modifiers

1. Static : belong to class
2. Final :
3. Abstract :

Final:

Final means cannot be changed.

It can be used with

1. Variable
2. Method
3. Class

Q: difference between final , finally, finalize()

Final - keyword

Finally - block

Finalize() --> G.C

Real example: final

Math formula : pi

Tax percentage

Abstract class:

An abstract class is class declared using abstract keyword.

Simple definition:

Abstract class : incomplete class that can contains **abstract** and normal method.

Note:

1. if any class have abstract method then we must use abstract keyword class.
2. We cannot create object of abstract class directly.

Interface: (1.7V)

Interface = contract that says what a class must do

Why abstract class is not 100% abstraction and why interface classes 100% abstraction ?